3 **(a)** PQRS comprises of a rough slope PQR of height 1.1 m, and smooth horizontal surface RS. PQR of total surface length 2.7 m is made up of a straight section PQ and curved section QR as shown in Fig. 3.1.

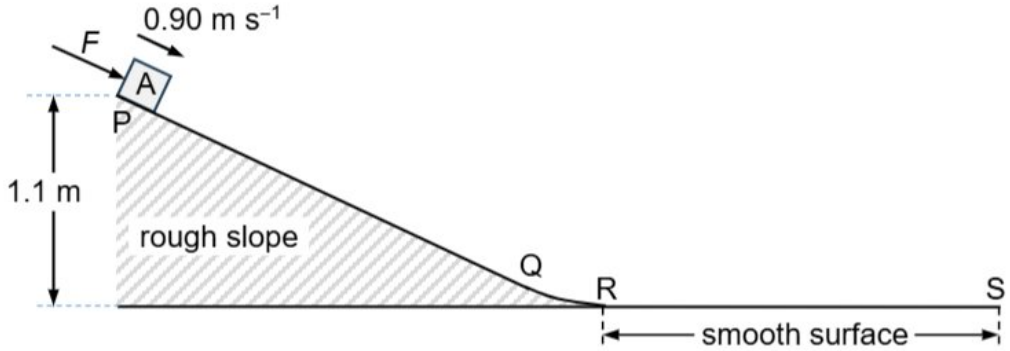


Fig. 3.1

Block A of mass 2.0 kg has an initial speed of 0.90 m s^{-1} at point P on rough slope. A force *F* is applied to block A as it moves along PQR. The magnitude of the *F* is 11 N and its direction is the same as the object's velocity at any instant along surface PQR.

The frictional force between the block A and the surface PQR may be taken to be constant at 6.4 N.

(i) Determine the speed of the block when it reaches R.

speed = ... m s^{-1} [3]

(ii) Use Newton's second law to explain how the speed of block A varies along PQRS.

.....

.....

.....

.....

.....

.....[3]

(b) Block X of mass 0.27 kg moves with a speed 0.67 m s^{-1} on a smooth horizontal surface. Block Y of mass 0.34 kg moves with speed 0.46 m s^{-1} in the opposite direction. A light spring of natural length 13.0 cm and force constant 120 N m^{-1} is attached to block Y as shown in Fig. 3.2.

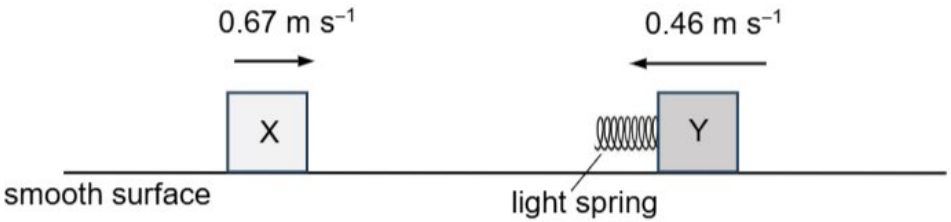


Fig. 3.2

When block X collides elastically head-on with block Y, there will be an instant when the spring experiences maximum compression.

Determine the closest distance of approach of the two blocks.

closest distance of approach = ... m [4]

[Total: 10]